Basis Data

Week 2 - DBMS

Data Model

- **Data Model** : A set of concepts to describe the structure of a database, and certain constraints that the database should obey.
- **Data Model Operations**: Operations for specifying database retrievals and updates by referring to the concepts of the data model. Operations on the data model may include basic operations and user-defined operations.

Categories of Data Model

1. **Conceptual (high-level, semantic) data models**: Provide concepts that are close to the way many users perceive data. (Also called entity-based or object-based data models.)
2. **Physical (low-level, internal) data models**: Provide concepts that describe details of how data is stored in the computer. Examples database configuration for to optimize data access speed.
3. **Logical data models:** It serves as an intermediary between the Conceptual Data Model (high-level design) and the Physical Data Model (implementation in a DBMS).

Schema vs Instances

- **Database Schema:** The description of a database. Includes descriptions of the database structure and the constraints that should hold on the database.
- **Schema Diagram:** A diagrammatic display of (some aspects of) a database schema.
- **Schema Construct:** A component of the schema or an object within the schema, e.g., STUDENT, COURSE.
- **Database Instance:** The actual data stored in a database at a particular moment in time. Also called database state (or occurrence).

Schema Diagram

Figure 2.1
Schema diagram for the database in Figure 1.2.

STUDENT

Name	Student_number	Class	Major

COURSE

Course_name	Course_number	Credit_hours	Department

PREREQUISITE

Course_number	Prerequisite_number

SECTION

Section_identifier	Course_number	Semester	Year	Instructor

GRADE_REPORT

Student_number	Section_identifier	Grade

Database Management System (DBMS)

A software system that enables users to define, create, maintain, and control access to the database.

Database Management System (DBMS)

DBMS provides the following facilities:

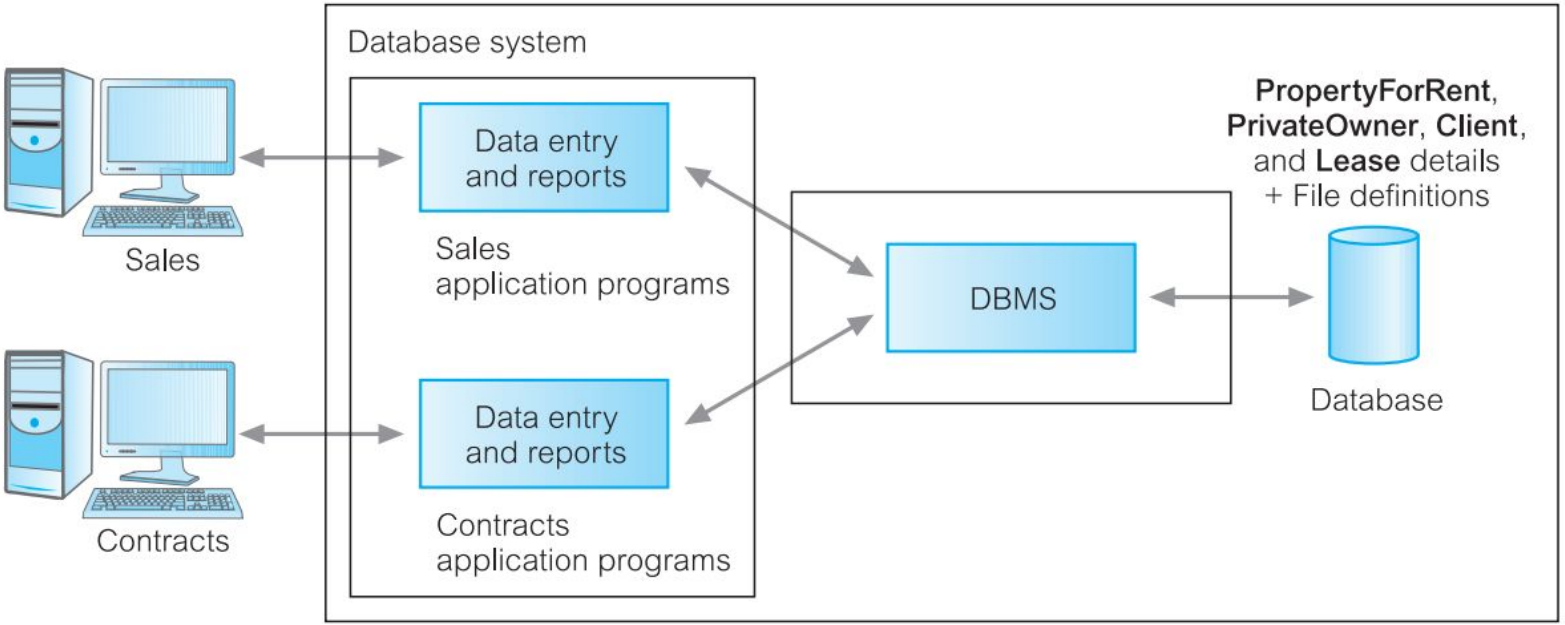
- It allows users to define the database, usually through a Data Definition Language (DDL). The DDL allows users to specify the data types and structures and the constraints on the data to be stored in the database
- It allows users to insert, update, delete, and retrieve data from the database, usually through a Data Manipulation Language (DML)
- It provides controlled access to the database (a security system, an integrity system, a concurrency control system, a recovery control system, and a user-accessible catalog)

Database Application Program

A computer program that interacts with the database by issuing an appropriate request (typically an SQL statement) to the DBMS

- Users interact with the database through a number of application programs that are used to create and maintain the database and to generate information
- The application programs may be written in a programming language or in higher-level fourth-generation language.

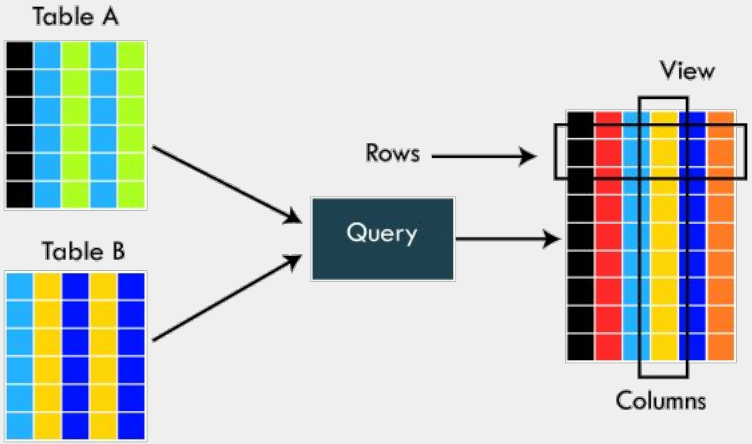
Database Processing



PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)
Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentFinish)

Views

- End-users are not too interested in how complex or easy a task is for the system
- DBMS provides another facility known as a view mechanism, which allows each user to have his or her own view of the database



Source : https://medium.com/@mohamad.ashour203/how-sql-views-are-important-in-the-role-of-data-analysis-a5e0b74b25ec

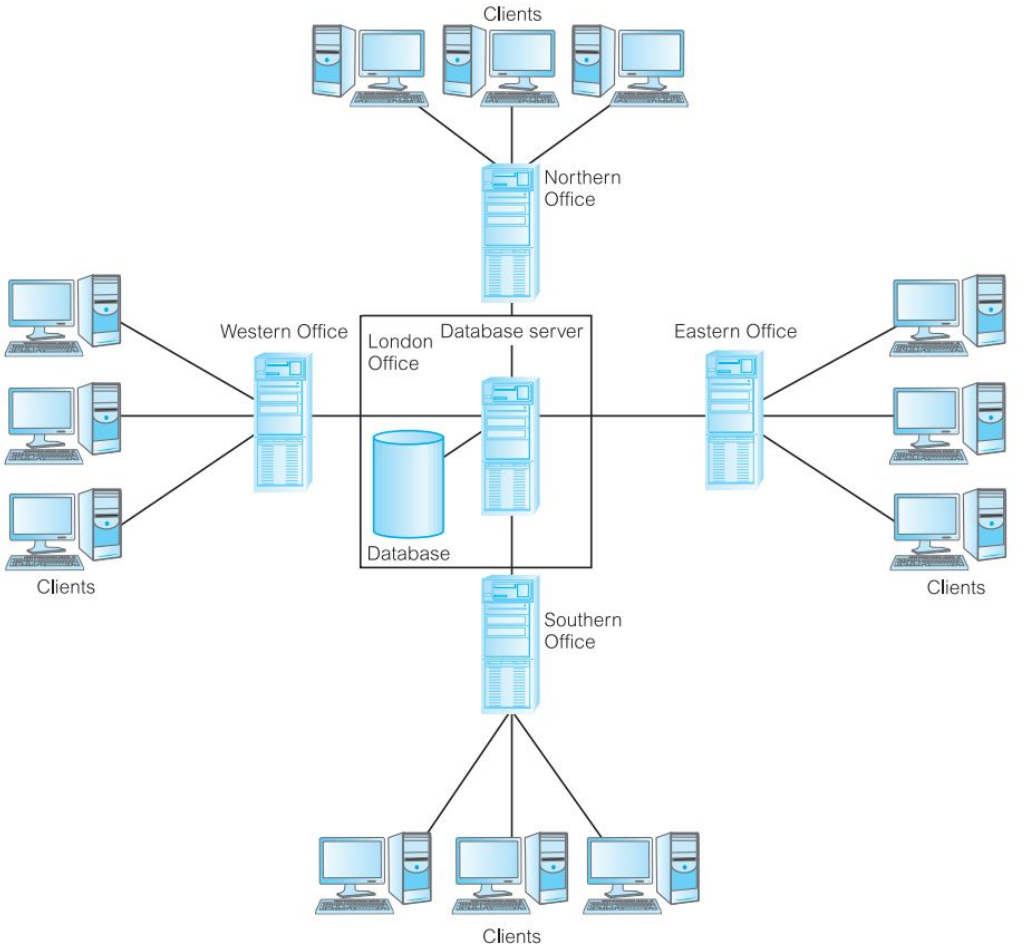
For example, we could set up a view that allows the Contracts Department to see only the data that they want to see for rental properties

Hardware

- The hardware can range from a single personal computer to a single mainframe or a network of computers
- The particular hardware depends on the organization's requirements and the DBMS used
  - Some DBMSs run only on particular hardware or operating systems, while others run on a wide variety of hardware and operating systems

This is called a client–server architecture:

- the backend is the server
- the frontends are the clients



People

- **Data administrators (DA):** responsible for the management of the data resource
  - Database planning
  - development and maintenance of standards, policies and procedures
- **Database administrators (DBA):**
  - physical realization of the database
  - physical database design and implementation,
  - security and integrity control,
  - maintenance of the operational system,
  - ensuring satisfactory performance of the applications for users

People

- **Database Designers**: logical database designers and physical database designers
- **Logical database designers** concerned with identifying the data (entities and attributes), the relationships between the data, and the constraints on the data that is to be stored in the database. Must have a thorough and complete understanding of the organization's data and any constraints on this data (the constraints are sometimes called business rules)
- Examples: a member of staff cannot manage more than 100 properties for rent or sale at the same time; member of staff cannot handle the sale or rent of his or her own property;

People

- **Physical database designer** decides how the logical database design is to be physically realized
  - mapping the logical database design into a set of tables and integrity constraints
  - selecting specific storage structures and access methods for the data to achieve good performance
- **Application developers**:
  - Once the database has been implemented, the application programs that provide the required functionality for the end-users must be implemented

Referensi

- Connolly, Thomas dan Begg, Carolyn. Database Systems: A Practical Approach to Design, Implementation, and Management 6th edition. Pearson. 2015.
- Elmasri, Navathe. Fundamental of Database Systems 7th edition. Pearson. 2016
- https://medium.com/@mohamad.ashour203/how-sql-views-are-important-in-the-role-of-data-analysis-a5e0b74b25ec